Engineering reference: governed manufacturing reporting

A sanitized technical case study covering architecture, contracts, state, validation, failure handling, tradeoffs, and impact methodology.

Reading guide

QUESTION	SECTION
How is the platform used?	1. Operational contract
What is implemented?	2. Architecture and data model
Why is delivery reliable?	3. State, validation, and recovery
How is evidence tested?	4. Verification matrix
How does the Reporting Assistant work?	5. Reporting Assistant internals
How are impact figures qualified?	6. Impact methodology

Public/private boundary

This reference uses synthetic data and sanitized topology. Tenant identifiers, employee data, production URLs, credentials, exact commands, permissions, and successor procedures remain in the private handoff.

1. Operational contract

The platform prepares decision-support evidence; people retain operational authority.

OUTPUT	PRIMARY USER	PLATFORM RESPONSIBILITY	HUMAN RESPONSIBILITY
Shift Draft	Supervisor	Collect, reconcile, organize, prepare spreadsheet	Review exceptions, add context, finalize
Daily Production	Manager	Combine production, molding, labor, quality, plan, downtime	Investigate constraints and prioritize follow-up
Weekly Production	Managers / teams	Create comparable period and trend views	Plan capacity and labor; prioritize improvement
Weekly Downtime	Managers / maintenance	Classify recurring and isolated downtime evidence	Set maintenance and coordination priorities

Decision boundary

Automation may prepare, classify, validate, refuse, and report state. It does not autonomously change staffing, production plans, maintenance priorities, or cross-department actions.

2. Architecture and data model

LAYER	COMPONENTS	RESPONSIBILITY
Intake	SharePoint, Forms, five Power Automate flows, machine events	Retain operational inputs and trigger bounded collection
Reporting core	Python ingestion/reconciliation, PostgreSQL	Normalize evidence and maintain source-linked serving state
Outputs	Workbooks, SharePoint libraries, API/SPFx dashboard	Expose prepared shift, daily, weekly, and live-status views
Intelligence	Chatbot, report doctor, governed oracle	Retrieve, diagnose, or validate without acquiring decision authority

Eight source-linked reporting tables

The serving model separates hourly production, quality, labor, molding, machine notes/events, final reports, report state, and lineage/settlement evidence. Separation prevents an output-friendly aggregate from erasing source identity or missingness.

CONTRACT	WHY EXPLICIT
Report identity	Prevents one period/family from settling another
Material identity	Distinguishes governed fact changes from capture-only metadata
Source lineage	Connects each prepared value to qualified evidence
Semantic certification	Tests meaning rather than file appearance alone
Publication/readback state	Separates accepted write from verified remote content

3. State, validation, and recovery

CANDIDATE	VALIDATED	PUBLISHED	READ BACK	SETTLED

State advances only when the evidence required for that transition exists. Missing evidence remains missing; an observed zero remains zero. A successful request is not equivalent to a verified remote artifact.

Failure scenarios

SCENARIO	CLASSIFICATION	SAFE BEHAVIOR	AUTHORITY
Required source absent	pending_source	Retain expected output; do not publish	No write
Equivalent replay	exact_no_op	Conserve material identity and remote version	No write
Write accepted; response lost	readback_pending	GET exact item; never repeat PUT	GET only
Contract or lineage drift	contract_refusal	Stop interpretation/publication	No write
Eligible late source	bounded_recovery	Advance same output identity once after validation	One qualified write

Release and rollback

One-writer releases serialize ownership and prevent overlapping publication. Promotion is guarded by output-suppressed/current-data checks. Exact rollback restores the last known-good immutable revision rather than reconstructing a deployment from mutable state.

4. Verification matrix

LAYER	REPRESENTATIVE CHECK	FAILURE PREVENTED
Source facts	Conservation and reconciliation against qualified inputs	Dropped or duplicated evidence
Workbook package	Visible sheets, formulas, structure, stored values	Client-dependent or malformed output
Semantics	Missing-versus-zero and family-specific rules	Meaning drift
Normalized rendering	Stable content independent of packaging noise	False changes from metadata
Remote readback	Exact item, version, ETag, content/material hash	Accepted-write ambiguity
Replay	Immediate equivalent cycle makes zero second PUT	Duplicate versions
Release	Serialized handoff, readiness guard, exact rollback	Concurrent writers / unsafe promotion

Claim discipline

Validation coverage proves the tested invariants and scenarios. It is not presented as blanket accuracy. Public evidence is sanitized; production-specific receipts and exact successor tests remain private.

Selected tradeoffs

CHOICE	BENEFIT	COST
Fail closed	No silent advancement on ambiguity	Some reports settle later
One writer	Clear ownership and recovery semantics	Serialized promotion
Same-item advancement	Stable operational identity and links	Requires strict replay/material checks
Bounded repair	Recovers only evidence-supported cases	Some historical cases remain terminal holds

5. Reporting Assistant internals

Publicly, these capabilities appear as one Reporting Assistant. Internally, three controlled subsystems retain distinct authority and failure behavior.

SUBSYSTEM	PURPOSE	BOUNDARY	REPRESENTATIVE REFUSAL
Chatbot	Source-grounded report retrieval	Structured response, validated citation, deterministic fallback	Unsupported question or missing source
Report doctor	Expected-output, exception, and settlement diagnostics	Read-only; no repair or publication authority	Non-admin or unavailable exact-period evidence
Governed oracle	Contract-bound interpretation and validation	Identity, lineage, state, semantics must agree	Stale, missing, or ambiguous evidence

Why three internal subsystems

Retrieval, diagnosis, and validation have different authority and failure modes. Keeping them separate makes refusal behavior understandable without presenting three competing public products.

Human control

The Reporting Assistant makes evidence easier to locate and inspect. Supervisors and managers still interpret context and choose operational action.

6. Impact methodology

MODEL	ASSUMPTIONS	CALCULATION	QUALIFICATION
Labor	20 supervisor + 10 manager hours/week; 52 weeks; documented role-rate assumptions	30 hours/week; about $52K/year	Modeled value, not booked savings
Retrieval	Representative 20-30 minute search; 1-2 minute bounded retrieval	About 90% reduction	Estimated, scenario-dependent
Downtime	5-minute response improvement; 1,000 widgets/hour/machine; 52.5 events/week; 52 weeks; $0.90/widget	Up to 227,500 widgets; about $205K contribution margin	Upper-bound opportunity, not realized production or margin

Limitations

The models depend on representative time studies, event frequency, throughput, response improvement, role rates, and contribution margin. They are useful for prioritization and sensitivity analysis, not guarantees. The synthetic workbook exposes editable inputs and formula-driven outputs so assumptions can be challenged directly.

Implementation references

The public repository contains sanitized architecture, state-machine documentation, tests, synthetic scenarios, release chronology, and claim boundaries. The private successor package contains exact dependencies, permissions, operating procedures, rollback, credentials transfer, and acceptance checks.